SIGSIM PADS 2024

The 38th International Conference on Principles of Advanced Discrete Simulation

ACM SIGSIM PADS 2024

Proceedings of the 38th ACM SIGSIM International Conference on Principles of Advanced Discrete Simulation

ACM SIGSIM-PADS'23
June 24-26, 2024, Atlanta, GA, USA

Conference Chair: Prof. Margaret Loper
Program Chair: Prof. Alessandro Pellegrini
Sponsor(s): ACM Special Interest Group on Simulation and Modeling (SIGSIM)
National Science Foundation (NSF)





The Association for Computing Machinery
1601 Broadway, 10th Floor
New York, New York 10019, USA

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ACM ISBN: 979-8-4007-0363-8

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Message from the Conference Chairs

Dear friends, Dear colleagues, Dear participants,

Welcome to the 38th ACM SIGSIM International Conference on Principles of Advanced Discrete Simulation (PADS). With a long tradition, ACM SIGSIM PADS has become a successful series of conferences that serves as a large international forum for presenting and sharing recent research results and technological developments in advanced simulation.

First and foremost, we thank *Philipp Andelfinger* and *Kevin Jin*, who have relentlessly helped the organizing committee in all aspects of the conference without taking on any specific role. Your help has been invaluable in shaping PADS '24.

This year's edition features a lively, engaging, and stimulating program with many opportunities for discussing new results, ongoing projects, and future trends in our field. The Technical Program Committee, consisting of 37 distinguished experts, selected the papers during a lively annual meeting. We sincerely thank the Technical Program Committee and the four external reviewers who helped in the review process for their invaluable effort in this process, which made this year's excellent program possible.

As in previous editions, the conference is organized in thematic sessions that reflect the diversity of topics related to modeling and simulation. This year, we received 33 submissions and accepted 11 as full papers, a selective rate of acceptance of 33%. Additionally, 6 contributions have been accepted as short papers to allow for a more dynamic and interactive event.

As usual, PADS strives to innovate, and this edition brings many novelties to increase the quality and significance of the conference. For the first time, we have introduced **two separate calls for papers**. In the first call, several authors received valuable comments on their research works, which allowed them to resubmit a revised version of the paper. The increased work by the Technical Program Committee has been beneficial in this regard.

Traditionally, PADS has selected a few outstanding papers for inclusion in a TOMACS Special Issue. The publishing of extended articles in the journal spanned over a year. To expedite the dissemination of new findings, this year's **TOMACS Special Issue** has been released at the start of the conference. The Technical Program Committee identified four submitted papers of outstanding quality, which have been promptly published in the special issue. We thank *Wentong Cai*, the TOMACS Editor in Chief, and all the ACM TOMACS Staff, particularly *Gita Delsing*, for their support.

Another innovation this year is the introduction of the **Test of Time Award**, which recognizes research that has demonstrated significant and sustained impact in our field over an extended period. A dedicated committee composed of *Philipp Andelfinger*, *Kevin Jin*, *Margaret Loper, Ernie Page, and Alessandro Pellegrini* selected one paper from PADS '04 and one from PADS '14. This year, the Test of Time Award goes to:

- *Jan Himmelspach* and *Adelinde M. Uhrmacher*. A component-based simulation layer for JAMES (from PADS 2004);
- *Jereme Lamps*, *David M. Nicol*, and *Matthew Caesar*. TimeKeeper: a lightweight virtual time system for Linux (from PADS 2014).

We express our appreciation to the **keynote speakers** for this year: *Georgia Tourassi*, Oak Ridge National Laboratory, Oak Ridge, TN, USA, and *Enlu Zhou*, Georgia Institute of Technology, Atlanta, GA, USA.

Enlu Zhou will present the application of data-driven simulation optimization within the context of digital twins, emphasizing the challenges of synchronizing real systems with their digital counterparts and adapting simulation models to accommodate real-time data for decision-making. This approach is increasingly crucial as digital twins become prevalent in various sectors, necessitating novel strategies for simulation and optimization based on real-time data analytics.

Georgia Tourassi will discuss the pioneering achievements and ongoing challenges of exascale computing at the Oak Ridge National Laboratory. Her keynote will explore how exascale computing capabilities, particularly the Frontier supercomputer, have enabled significant advancements in generative AI and large-scale scientific simulations. Tourassi will address the critical balance between the computational power provided by such technologies and the constraints imposed by energy consumption, outlining strategies to optimize energy efficiency while continuing to push the boundaries of computational science and its applications. Together, these keynotes will offer comprehensive insights into the cutting-edge of computational technology and its application in solving complex, large-scale problems.

We also express our gratitude to the **Best Paper Award** Committee, *Adelinde Uhrmacher*, *Wentong Cai*, and *Francesco Quaglia*, for their diligent work in reviewing and selecting the best paper awards. The candidates short-listed for the best paper award for this year are (in no particular order):

- *Juri Tomak* and *Sergei Gorlatch*. A Toolset for Predicting Performance of Legacy Real-Time Software Based on the RAST Approach;
- *Srikanth Yoginath*, *Pratishtha Shukla*, *Sudip Seal* and *James Nutaro*. SpecSims: A Scalable Speculative Tree-based Simulation Cloning Framework For Finite Memory Machines.

The best paper award winner from the candidates mentioned above will be announced during the social dinner. In keeping with tradition, we announce the best paper award winners of the previous edition of PADS'23:

- *Philipp Andelfinger*, *Till Köster*, and *Adelinde Uhrmacher*. Zero Lookahead? Zero Problem. The Window Racer Algorithm (full);
- *Elkin Cruz-Camacho*, *Kevin A. Brown*, *Xin Wang*, *Xiongxiao Xu*, *Kai Shu*, *Zhiling Lan*, *Robert B. Ross*, *Christopher D. Carothers*. Hybrid PDES Simulation of HPC Networks Using Zombie Packets (short).

We continue the tradition started in 2018 of allowing the authors of original papers accepted to ACM TOMACS, the journal of our Special Interest Group, to present their work at the conference. This year, the **TOMACS-to-PADS initiative** will feature presentations by *Xiaoliang Wu, Alexander Kolar, Joaquin Chung, Dong Jin, Martin Suchara,* and *Rajkumar Kettimuthu*. As usual, it is a great opportunity to disseminate consolidated research results within the PADS community.

A total of 13 extended abstracts were submitted to the PhD Colloquium, of which 11 were accepted based on relevance and technical soundness. The Colloquium will allow early-stage researchers to connect with the community and obtain feedback on their work.

PADS continues actively participating in ACM's **Artifact Evaluation** initiative, strengthening the trust in published results within the modeling and simulation field.

In total, artifacts for 6 of the accepted papers were submitted for evaluation. After careful evaluation, the reproducibility committee was able to award multiple badges to all papers that opted in the initiative. We extend our sincere gratitude to the authors and the reproducibility committee for making this valuable contribution to the community.

The success of a conference is mainly on the part of the participants; we therefore warmly thank all the contributors and participants. The team of chairpersons did a huge amount of work, which finally led to a very attractive program. We wish you a successful conference and a pleasant stay in Atlanta!

Margaret Loper
SIGSIM PADS '24 General Chair
Georgia Tech Research Institute

Alessandro Pellegrini
SIGSIM PADS '24 Program Chair
Tor Vergata University of Rome

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